Weekly Research Report — Week 4

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Summary of Work Done

A custom GraphRAG pipeline was implemented and evaluated, extending the comparison to five architectures. The Introduction was restructured following supervisor feedback and a Results section was added, which the draft had previously lacked.

GraphRAG implementation

Rather than adopt the Microsoft GraphRAG package wholesale, a custom pipeline was built to retain control over entity typing and retrieval:

1. **Entity and relation extraction** — `llama-3.3-70b-versatile` prompted for structured JSON output over MedQA training documents. Entity types: disease, symptom, medication, treatment, procedure. Relation types: treats, causes, indicates, contraindicates.
2. **Graph construction** — NetworkX, typed nodes and edges
3. **Retrieval** — query embedded with all-MiniLM-L6-v2, cosine similarity against node embeddings, top-k subgraph returned as formatted triples

Graph built from 50 training documents: **282 entities, 223 relationships.**

Extended architecture comparison

Five architectures evaluated on a 20-question MedQA sample:

Architecture	Accuracy
Baseline LLM	70%
Flat-vector RAG	65%
Static GraphRAG	70%
Agentic GraphRAG with self-correction	75%
Agentic GraphRAG with dynamic PubMed	75%

Findings against the hypotheses

- **H1** — flat-vector RAG falls 5 points below baseline, confirming the retrieval-noise hypothesis. Static GraphRAG recovers the baseline exactly, suggesting graph structure removes the noise penalty without yet adding value.
- **H2** — agentic self-correction lifts accuracy to 75%. Self-correction triggered on 10 of 20 questions under a confidence threshold of 6/10, with 80% accuracy on triggered questions against 70% on the remainder. The confidence signal identifies a more answerable subset.
- **H3** — dynamic PubMed integration matches but does not exceed the static agentic system. Two candidate explanations: the 100-character heuristic used to formulate the literature query is crude, and the self-correction loop may already absorb the benefit that dynamic expansion would otherwise provide.

Writing

Introduction restructured per supervisor comments: explicit problem statement, then the solution space developed across three separated paragraphs — flat-vector RAG and its retrieval-noise failure, static GraphRAG and its staleness failure, agentic GraphRAG as the convergent direction. Closes with three research questions and a forward pointer to findings.

Results section (Section V) added with the five-architecture table and per-hypothesis analysis.

Limitation identified

The sample is 20 questions. At that size a single question shifts accuracy by 5 percentage points, which is the entire margin between several architectures. No statistical claim can be supported. Scaling to the full 1,273-question test set is the priority.

Plan for Week 5

1. Address Dr. Fiaidhi’s draft comments
2. Scale evaluation beyond the 20-question pilot
3. Add statistical significance testing

Blockers

Scaling to the full test set at current API rates needs a decision on paid API access.